A Project Report on

CONTEXTUAL BANDIT-BASED HYBRID BIT AND SEMANTIC TRANSMISSION SELECTION FOR UAV-ASSISTED NETWORKS

submitted in partial fulfilment of the requirements for the course

MACHINE LEARNING
25SC2107E

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by

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DEPARTMENT OF **ELECTRONICS & COMMUNICATION ENGINEERING**

Name of Title: Contextual Bandit-Based Hybrid Bit and Semantic Transmission Selection for UAV-Assisted Networks

1. Abstract:

Unmanned Aerial Vehicle (UAV)-assisted wireless networks are increasingly relied upon for time-critical, resource-constrained applications such as disaster response, precision agriculture, and search-and-rescue, where air-to-ground channels fluctuate rapidly due to UAV mobility and intermittent line-of-sight. Conventional bit-level transmission guarantees exact reconstruction but is bandwidth- and energy-inefficient for content whose meaning survives approximate delivery, while emerging semantic communication techniques transmit only task-relevant meaning at the cost of added on-board inference latency and unpredictable degradation under poor channel conditions; a further compressed-semantic variant trades additional fidelity for even lower latency and energy use. Because no single transmission mode is optimal across all operating conditions, this project addresses the problem of adaptively selecting, for each arriving task, among conventional bit transmission, semantic transmission, and compressed semantic transmission, using four decision-relevant signals: instantaneous channel state, task urgency, available processing-delay budget, and the task's semantic quality requirement. We frame this decision as a contextual multi-armed bandit problem, in which a learning agent observes an evolving context vector at every task arrival and selects a transmission arm so as to maximize a reward that jointly captures task success, latency, and energy cost.

The proposed approach trains and compares contextual bandit algorithms — LinUCB and Thompson Sampling — as the primary learning method, with a Deep Q-Network trained on the same task stream as a secondary baseline to test whether explicitly modelling sequential dependencies, such as UAV battery depletion and queue backlog, offers meaningful gains over the lighter-weight bandit formulation. Both learners are benchmarked against fixed single-mode policies, random selection, and a hand-tuned threshold heuristic, within a simulated UAV air-to-ground environment that incorporates a probabilistic line-of-sight / non-line-of-sight channel model and a lightweight semantic-quality proxy. Performance is evaluated using cumulative reward, regret against an oracle policy, task success rate, end-to-end latency, energy consumption, and semantic quality of experience, together with ablation studies isolating the contribution of each context feature and sensitivity analyses across traffic load and channel volatility. We expect the contextual bandit agents to converge to a near-optimal, low-regret policy substantially faster than the DQN baseline, and the learned adaptive policy to outperform every fixed transmission mode once evaluated across heterogeneous urgency levels and channel conditions — demonstrating that dynamic, context-aware mode switching offers a practical, deployable improvement for UAV-assisted communication systems without requiring any redesign of the underlying semantic codec.

2. SOFTWARE REQUIREMENTS:

This project is implemented entirely in Python within the Google Colab environment, which provides a hosted notebook with pre-installed scientific-computing libraries and free GPU access, eliminating the need for any local installation. Core numerical operations, data logging, and visualization are handled using NumPy, SciPy, Pandas, and Matplotlib, while PyTorch, available by default with GPU support in Colab, is used to implement the Deep Q-Network baseline and the semantic-quality proxy model. The contextual bandit algorithms (LinUCB and Thompson Sampling) are implemented directly using NumPy, and Gymnasium and Stable-Baselines3 are installed via pip to provide a standardized reinforcement-learning environment interface and a verified DQN implementation, respectively. Model checkpoints and experiment results are stored persistently using Google Drive integration, and Git is used for version control to ensure the codebase remains organized and reproducible throughout development.

Given that Google Colab operates on a session-based cloud infrastructure, the training pipeline incorporates periodic checkpointing to Google Drive to safeguard against runtime disconnections, which can occur after prolonged inactivity or upon reaching the platform's maximum session duration. GPU acceleration is enabled through Colab's runtime configuration prior to training, substantially reducing the computation time required for the Deep Q-Network baseline compared to CPU-only execution. All dependencies beyond the pre-installed scientific stack are installed at the start of each session using pip, ensuring a consistent and reproducible software environment across runs without requiring dedicated local hardware or manual configuration. This browser-based, cloud-hosted workflow allows the complete experimental pipeline — from data simulation through model training and evaluation — to be executed and iterated upon entirely within a single, accessible development environment.

Signature of the Supervisor

Course Instructor